Rates and Unit Rates Flagship

Lesson 4-1-flagship

Name: ____________ Date: ____________ Class: ____________

ARCADE BUILDER MISSION

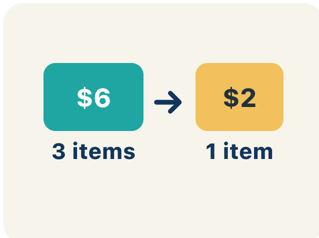
Best Booth in the Arcade

You are the new operations manager at NeonPlex Arcade. Two ticket booths are competing for players — Booth A charges $3 for 5 games, Booth B charges $5 for 8 games. Players want the best deal, and your job is to crown the winning booth by mastering unit rates.

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



$12 for 4 games is a rate

Rate

Write the definition:

____

____

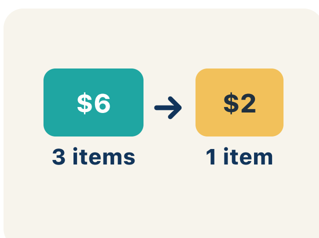
$3 per 1 game

Unit Rate

Write the definition:

____

____



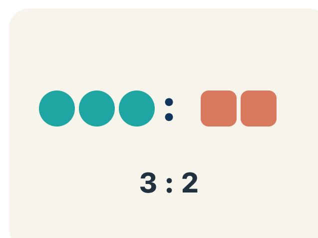
60 miles per hour means 60 miles for every 1 hour

Per

Write the definition:

____

____



5 to 3 or 5:3

Ratio

Write the definition:

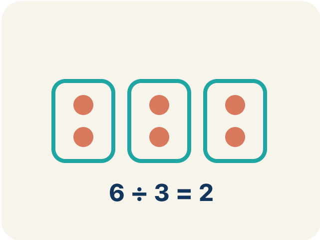
____

____

Pack A: $0.50 per pencil vs Pack B: $0.40 per pencil — Pack B is the better buy

Better Buy

Write the definition:



$12 ÷ 4 games = $3 per game

Divide

Write the definition:

Guided Notes

Level 2 Standard

🎯 **WHAT WE'RE LEARNING TODAY**

I can find a unit rate to compare prices and decide the better buy.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Rate | **Unit Rate** | **Per** | **Ratio** | **Better Buy** | **Divide**

1. A ratio that compares two quantities with different units is a ______________ .

2. A rate that tells the cost or amount for one unit is a ______________ .

3. The word that means 'for each one' is ______________ .

4. A comparison of two quantities is a ______________ .

5. The option with the lower unit rate, costing less per item, is the ______________ .

6. To find a unit rate, I ______________ the price by the number of units.

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: A booth charges $4.50 for 9 games. What is the unit rate?

A. $0.50 per game

B. $0.45 per game

C. $2.00 per game

D. $4.50 per game

Step 1 $4.50 ÷ 9 = $0.50 per game.

✅ **Answer:** A. $0.50 per game

🤝 We do — together

Solve this one with your class using the same steps.

Problem: A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

A. 30 miles per gallon

B. 6 miles per gallon

C. 180 miles per gallon

D. 36 miles per gallon

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ You do — your turn

Now try one on your own. Show every step.

Problem: A pack of 8 markers costs $6.00. What is the cost per marker?

A. $0.75

B. $0.80

C. $1.33

D. $0.60

Show your work:

____

____

____

Try It

Solve on your own. Check the answer key when you are done.

1. Lock 2 — Speed Door: Two racing games unlock the door. Pixel Racer scores 250 points in 5 minutes. Turbo Drift scores 180 points in 4 minutes. Which game earns points at the faster rate?

A. Pixel Racer (50 points per minute)

B. Turbo Drift (50 points per minute)

C. Pixel Racer (45 points per minute)

D. They earn points at the same rate

Show your work:

2. Lock 3 — Claw Machine Vault: Three coin deals glow on the screen. Which deal is the best value (lowest cost per play)?

A. $5.00 for 10 plays

B. $6.00 for 8 plays

C. $9.00 for 12 plays

D. $7.00 for 10 plays

Show your work:

Stretch Your Thinking Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

An arcade sells tokens in three packs: 25 tokens for $5, 60 tokens for $10.80, and 100 tokens for $19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.

Sentence starter: The student is ____ because the best deal is the ____ pack with a unit rate of $____ per token.

Show your work:

____

____

____

____

Reflect — Exit Ticket

A store sells 6 pencils for $1.50 and 10 pencils for $2.80. Which is the better buy?

A. 6 for $1.50 — $0.25 each

B. 10 for $2.80 — $0.28 each

C. They cost the same per pencil

D. Not enough information

Your answer:

____

____

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Answer Key & Teacher Guide

1. **Notes 1:** Rate
2. **Notes 2:** Unit Rate
3. **Notes 3:** Per
4. **Notes 4:** Ratio
5. **Notes 5:** Better Buy
6. **Notes 6:** Divide
7. **We Try (notes):** A. 30 miles per gallon *— 180 miles ÷ 6 gallons = 30 miles per gallon.*
8. **You Try (notes):** A. $0.75 *— $6.00 ÷ 8 = $0.75 per marker.*
9. **Try It 1:** A. Pixel Racer (50 points per minute) *— Pixel Racer: 250 ÷ 5 = 50 points per minute. Turbo Drift: 180 ÷ 4 = 45 points per minute. 50 > 45, so Pixel Racer is faster.*
10. **Try It 2:** A. $5.00 for 10 plays *— $5.00 ÷ 10 = $0.50 per play; $6.00 ÷ 8 = $0.75 per play; $9.00 ÷ 12 = $0.75 per play; $7.00 ÷ 10 = $0.70 per play. The lowest unit rate, $0.50 per play, is the best value.*
11. **Exit Ticket:** A. 6 for $1.50 — $0.25 each *— $1.50 ÷ 6 = $0.25 per pencil. $2.80 ÷ 10 = $0.28 per pencil. $0.25 < $0.28, so 6 for $1.50 is the better buy.*